Shuxin Cao

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EDUCATION

- **Georgia Institute of Technology** — Atlanta, USA
  Ph.D. in Robotics, Advised by Asst. Prof. Animesh Garg — *Aug 2022 - Present*
- **Nanyang Technological University** — Singapore
  Bachelor of Engineering (Electrical and Electronic Engineering) with a Minor in Mathematics — *Aug 2017 - Jun 2021*
  NTU Science and Engineering Undergraduate Scholarship

RESEARCH EXPERIENCE

- **Georgia Institute of Technology** — USA
  Advised by Asst. Prof. Animesh Garg — *Aug 2023 - Present*
  - **Generalized Imitation Learning**: Developed a keypoint-based policy framework for robotic manipulation that enables generalization across intra-category object variations in geometry, size, and orientation. Introduced a new dataset with diverse tool-use tasks and demonstrated that our approach significantly improves success rates on both seen and unseen instances.
  - **Hierarchical Spectral VLA**:
- **Nanyang Technological Universtiy** — Singapore
  Advised by Prof. Xie Lihua — *Jan 2019 - May 2020*
  - **Transfer Learning & IOT**: Proposed a novel model using cross-domain distillation to solve the domain adaptive problem of IoT edge devices. Conducted Experiments on WiFi gesture recognition under IOT environment using channel state informa- tion(CSI) data.
  - **Unsupervised domain adaptation**: (i) Formulated an imbalanced domain adaptation (IDA) problem where both label shift and co-variate shift occur between the two domains. (ii) Developed a novel cluster-level discrepancy minimization (CDM) to solve the problem. CDM proposes cross- domain similarity learning to learn tight and discriminative clusters, which are utilized for both feature-level and distribution-level discrepancy minimization, palliating the negative effect of label shift during domain transfer
- **The State Key Laboratory for Manufacturing Systems Engineering, XJTU** — Xi'an, China
  Advised by Prof. Gao Feng — *Jan 2017 - Jan 2018*
  - **Fault diagnosis and cloud monitoring system**: In order to solve the problem of fault diagnosis of the servo motor of CNC equipment, we proposed a fault diagnosis an algorithm based on dual-stack sparse auto-encoder models. By establishing the simulation model, we determined that this method can achieve a robust, high-accuracy fault recognition rate. We also developed a server status tracking system for the cooperating company to provide important insights

PROFESSIONAL EXPERIENCE

- **Nanyang Technological Universtiy** — Singapore
  Research Assistant (Full-Time), Satellite Research Centre, Collaborate with DSO National Laboratories — *Aug 2021 - Aug 2022*
  - **Transfer Optical image to SAR image**: **(i)**Optimizing EM solver SAR Simulator. Using EM solver to simulate SAR images from 3D models in order to assist models in learning the imaging principles of high-dimensional data.
    (ii) Exploit the advanced domain-aware GAN models for SAR-EO domain transfer and modality adaptation. Design a novel deep multichannel network to fully leverage the unique properties of SAR and EO imageries, by exploiting the cross-modality and cross-domain correlation for remote sensing applications.
- **Microsoft Research Asia** — Beijing, China
  Intern, Software Analysis Team, Advised by Dr. Lin Qingwei, Collaborate with Microsoft Redmond — *Dec 2019 - Jan 2021*
  - **Cloud system optimization**: **(i)** Designed and developed an Azure Compute Efficiency Tracker to systematically track and analyze all relevant aspects of Azure Compute's efficiency in an all-up manner.
    (ii) Optimize previous resource allocation algorithms to reduce operating failures and increase processing speed.
  - **Time series prediction & Fault detection**: **(i)** Propose a new algorithm for time series prediction that automatically recommends and ensemble models. Considering the nonlinear and uncertain factors, the algorithm determines trade-offs between the high accuracy of deep learning methods and the high real-time of traditional models.
    (ii) The algorithm was developed into a python library which had already been used in the research team and applied on the Azure platform. The library assists cloud fault detection and resource allocation, timely warning when the system does not meet the normal trend, thereby improving the robustness and efficiency of the system.

PUBLICATIONS

Yang, J., Yang, J., Wang, S., Cao, S., Zou, H., & Xie, L. (2021). Advancing imbalanced domain adaptation: Cluster-level discrepancy minimization with a comprehensive benchmark. *IEEE Transactions on Cybernetics.*

Qian, H., Zhang, P., Ji, S., Cao, S., & Xu, Y. (2021). Improving representation consistency with pairwise loss for masked face recognition. In *Proceedings of the ieee/cvf international conference on computer vision* (pp. 1462–1467).

Yang, J., Zou, H., Cao, S., Chen, Z., & Xie, L. (2020). Mobileda: Toward edge-domain adaptation. *IEEE Internet of Things Journal*, *7*(8), 6909–6918.

Zhang, Z., Cao, S., & Cao, J. (2018). fault diagnosis of servo drive system of cnc machine based on deep learning. In *2018 chinese automation congress (cac)* (p. 1873-1877). doi: 10.1109/CAC.2018.8623472

Cao, S., Wang, L., & Garg, A. (2026). Keygen: Unsupervised keypoint-based object-centric representations for category-level policy generalization. *in submission to IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS).* (under review)

Cao, S., Wang, L., Byrnes, W., Chen, Y., Du, Y., & Garg, A. (2026). Hierarchical policy learning via spectral decomposition. In *Icml 2026.*

Chen, Y., Jian, Y., Dong, X., Cao, S., Wu, J., Vela, P., … Chen, D. (2026). Vista: Enhancing visual conditioning via track-following preference optimization in vision-language-action models. *arXiv preprint arXiv:2602.05049.*